

Problem 1. For the learning problems described below, answer the following questions:

- (i) is the problem a regression or a classification?
 - (ii) is the problem supervised or unsupervised?
 - (iii) if available, what is the number of features p and the number of data points N ?
1. Consider the problem of predicting the electric energy consumption in a city at 9 PM tomorrow using the values of the energy consumption at 9 PM on the previous 21 days. To train your predictor, you have access to historical data of the past 10 years. (Hint: Your answer about N should take into account that the initial 21 days of this 10-year period cannot be used as outputs in your training set.)
 2. Consider the problem of detecting a failure in a power grid; in particular, you want to find out whether a particular transmission line has failed or not at a given time. To make a decision, you use voltage measurements in 10 different nodes of the power grid and use voltage fluctuations to decide if the transmission line is operational or if it failed. These voltages are measured using a device able to sample these voltages at a temporal resolution of 600Hz. Your algorithm needs to make a decision about failures using the voltage measurements corresponding to the last 5 seconds. You have access to historical data containing 100 examples of time series with and without line failures.

Problem 2. For each one of the random variables below, plot their pdf (or pmf), and compute their expectations, as well as their variances:

1. The outcome of the flip of a fair coin, where 'Head'=1 and 'Tail'=0.
2. The sum of three fair coins flipped simultaneously.
3. The outcome of flipping a fair die with 4 faces.
4. The sum of a fair die with four faces and a coin ('Head'=1 and 'Tail'=0) thrown simultaneously (and independently).
5. A uniform (real) number in the range $[-1, 1]$.
6. The sum of a fair coin and a real number chosen uniformly at random in the range $[-1, 1]$.

Problem 3. Throw a fair dice with 6 faces and denote its outcome by the random variable $D \in \{1, 2, \dots, 6\}$. Then, draw a random value U uniformly at random from the interval $[0, d]$, where d is the value of the dice you previously threw. Answer the following questions:

1. Compute the marginal probabilities $\Pr(D = d)$ for $d \in \{1, 2, \dots, 6\}$ and $\Pr(U \in [0, 1])$. Hint: you might

want to use the total probability theorem; namely $\Pr(U \in [0, 1]) = \sum_{d=1}^6 \Pr(U \in [0, 1] | D = d) \Pr(D = d)$

2. Compute the marginal expectation $E(U)$. Hint: you might want to apply the following identity,

$$E(U) = \sum_{d=1}^6 E(U | D = d) P_D(d)$$